Ryan Zernach Full-Stack AI/ML Software Engineer

Ryan@Zernach.com • Ryan.Zernach.com • LinkedIn • Github • StarLink • Remote

HIGHLIGHTS

- Distributed systems engineer with 10+ years of experience in Mobile Apps, Agents, MLOps, & Cloud Infrastructure
- Shipped end-to-end CellarChat™ chatbot feature to 1M+ members, complete with CI/CD LLM evaluation pipelines
- Designed distributed backend systems that support scalable AI-powered eCommerce VoIP automation workflows
- Boosted mobile app ratings to 4.9★ (10k+ reviews) & lifted recurring revenue by 25%+ w/ Apple Pay subscriptions
- New users signup everyday for the apps that I've created & marketed as solutions to my own personal problems
- Awarded Top 10 of 10,000 international projects in 2024 NASA Space Apps hackathon with AI Voice Assistant

TECH STACK

- LLMOps / RAG / MCP / Agents / VoIP / Evaluations
- OpenAPI / LangChain / Neo4j / Semantic Kernel
- Machine Learning Pipelines / Computer Vision
- Python / PyTorch / PySpark / ETL / TensorFlow
- Cloudflare / AWS / Azure / GCP / CloudOps
- DevOps / Docker / Kubernetes / Terraform
- Vector Databases / SQL / GraphQL / NoSQL
- Git / Bash / Cursor / Claude / Orchestration
- AuthN / AuthZ / OAuth / Identity / Security
- React Native / iOS / Android / Swift / Kotlin / Expo
- TypeScript / React / Redux / Node / Next / Streamlit
- WebSockets / Queues / RabbitMQ / Redis / WebRTC
- FastAPI / Express / Rust / .NET / C# / Go / Java / Ruby
- Blockchain / dApps / Wallets / Ethereum / Solana
- Apple Pay / Subscriptions / Stripe / IAP / Webhooks
- SEO / Marketing Automations / Campaign Attribution
- IMAP / SMTP / Emails / SMS / Twilio / Push Notifications
- Systems Observability / Analytics / Admin Dashboards

EXPERIENCE

AI Engineer / Mobile Engineer (Full-Stack) • **CellarTracker** • *Remote • StarLink* ***Aug 2023 - Jul 2025***

- Wrote CellarChat™ LLM chatbot for 1M+ active members using TypeScript front-end & Python back-end
- Architected OpenAPI tool-calling pipeline, fine-tuning pipeline, LLM evaluations, & observability systems
- Integrated intelligent ML-powered recommendations for world's largest online database of wine reviews
- Maintained Microsoft Azure infrastructure & CI/CD pipelines used by international engineering members
- Helped to boost app store ratings to 4.9★ average with 10,000+ verified 5-star ratings on iOS and Android
- Embedded Apple Pay subscriptions in the React Native app to generate additional 25%+ recurring revenue
- Improved user's experience of cellar management CRUD workflows to improve consistency and reduce taps
- Implemented animations, barcode scanning, activity feed posts, support ticket refactors, & OCR image search
- Streamlined the Redux store for fast, efficient, and persistent global state management between app sessions
- Rebuilt the React Native app's analytics implementations for tracking critical user interactions and app behavior

Mobile Engineer (Full-Stack) • **Archlife Industries Software** • *Remote • Miami* ***Sept 2021 - Aug 2023***

- Built eTrading app to compete with Robinhood by exchanging stocks/crypto & securely connecting bank accounts
- Engineered companion admin web dashboard to monitor mobile app performance & AWS infrastructure health
- Constructed core mobile app functionality, including: copy portfolio, trending investors, & suggested stocks
- Built social network features, such as direct chat, user profiles, posts, likes, emojis, & threaded comments
- Wrote the data-visualization logic for viewing and comparing the performance of different stocks/crypto
- Iterated on group chat features, push notifications, & native calendar sync to improve user satisfaction
- Evaluated Google Crashlytics to reduce edge case crashes and improve p95 stability and performance
- Transitioned eSports mobile app codebase from Javascript to Typescript for long-term maintainability
- Improved computational and network efficiencies by reducing load times and runs to the database
- Implemented updates to front-end React Native according to the design team's mockups in Figma
- Remedied bluetooth connection & map performance for React Native telecommunications app

Data Science & Machine Learning Tutor • **edX** • *Vanderbilt, UC Berkeley, +24 others* ***July 2020 – Sept 2021***

- During pandemic, taught 900+ online college students Python, machine learning, TypeScript, websites, & databases
- Managed busy schedule of remotely teaching 30+ university students per week in 1:1 setting (one hour per person)
- Dozens of universities were subscribed to my employer's curriculum and online tutoring resources for students

React Web Developer (Full-Stack) • **Georgia Hybrid Landscaping** • *Atlanta, GA* ***Oct 2015 – Jan 2019***

- Engineered user experience of uploading videos/images, survey questions, & receiving PDF pricing proposal
- Converted PHP websites to React and marketed businesses to generate $500k+ of gross sales in 3 years
- Configured customized outbound marketing email campaign schedulers & listeners using IMAP/POP
- Setup custom observability dashboards for marketing, analytics, API logs, & database admin

EDUCATION

Computer Science & Mathematics • **Georgia Southern University** • *Statesboro, GA* ***2012***

- Visual Basic, Calculus, C, C++, R, Statistics, Data Structures, Programming Principles, Linear Regression

Turfgrass & Landscape Management • **University of Georgia (NALP)** • *Athens, GA* ***2015***

- Design Principles, Science of Grass, Outdoor Business Management, Landscape Architecture

Data Science & Machine Learning Bootcamp • **Lambda School** • *Y Combinator* • *Remote* ***2020***

- Recommendation Engines, Computer Vision (CV) Models, Time-Series Forecasting, Natural-Language Processing
- Python, SciKit-Learn, PySpark, Data Engineering, PostgreSQL Databases, Back-End APIs (Flask / FastAPI)

AI Engineering Fellowship (G3) • **Gauntlet AI** • *Remote* • *Austin, TX* ***2025***

- Mastered how to command agents, automate workflows, and 100x enterprise-grade software engineering
- Harnessed the power of RAG, multi-agent systems, OpenAPI, and MCP to build agentic automations

CERTIFICATIONS

- **2025**: *Gauntlet AI Engineering Fellowship (G3)*
- **2024**: *NASA Space Apps Hackathon Finalist*
- **2022**: *Ethereum Blockchain Developer*
- **2020**: *TensorFlow Computer Vision*
- **2020**: *AutoML Pipelines (DataRobot Academy)*
- **2019**: *Google Cloud Platform GCP Developer*
- **2019**: *AWS Microservices Practitioner*
- **2019**: *Y Combinator Startup School*

Thank you for reviewing my resume!
Ready to interview me? Let's chat:

Schedule Meeting